Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Full Stack development MERN Practical
Practical- 8

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Class: TYIT	Batch:1
Date of Assignment: 30-09-2026	Date/Time of Submission: 30-09-2026

Practical- 8

Aim: API Creation and Integration

A. Create a simple REST API using Express

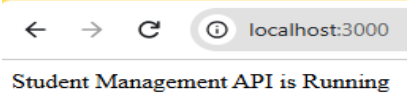
- Create a folder & open that in vs code
- Run: npm init -y to create package.json file.
- Add server.js file
- Run command: npm install mongoose express cors

Code: server.js:

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
const app = express();
const PORT = 3000;
// Middleware
app.use(cors());
app.use(express.json());
// Connect MongoDB
mongoose.connect("mongodb://127.0.0.1:2701
7/CollegeDB") .then(() => {
 console.log("MongoDB Connected
Successfully"); })
 .catch((error) => {
 console.log("MongoDB Connection Failed");
 console.log(error);
 });
// Simple REST API
app.get("/", (req, res) => {
 res.send("Student Management API is
Running"); });
app.listen(PORT, () => {
 console.log(`Server is running on
http://localhost:${PORT}`); });
```

Output:

```
C:\Program Files\nodejs\node.exe .\server.js
Server is running on http://localhost:3000
MongoDB Connected Successfully
```



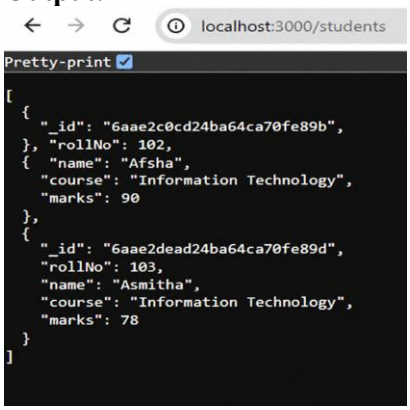
B. Implement a GET API to retrieve data.

Code: server.js:

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
const app = express();
const PORT = 3000;
// Middleware
app.use(cors());
app.use(express.json());
// Connect MongoDB
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 console.log("MongoDB Connection Failed");
 console.log(error);
 });
// Simple REST API
app.get("/", (req, res) => {
 res.send("Student Management API is
Running"); });
 app.get("/students", async (req, res) => {
 try {
 const students = await
mongoose.connection.db
 .collection("students")
 .find()
 .toArray();
 res.json(students);
 }
 catch (error) {
 res.status(500).json({
 message: "Error fetching students"
 });
 }
});
app.listen(PORT, () => {
```

```
 console.log(`Server is running on
http://localhost:${PORT}`); });
```

Output:



C. Fetch data from the API in a React application. (Library: npm install axios)

Code: App.jsx:

```
import { useEffect, useState } from "react";
import axios from "axios";
function App() {
const [students, setStudents] = useState([]);
useEffect(() => {
axios.get("http://localhost:3000/students")
.then((response) =>
{setStudents(response.data);})
.catch((error) => {console.log(error);}); }, []);
return (
<div>
<h1>Student List</h1>
<table border="1">
<thead>
<tr>
<th>Roll No</th>
<th>Name</th>
<th>Course</th>
<th>Marks</th>
</tr>
</thead>
<tbody>
{students.map((student) => (
<tr key={student._id}>
<td>{student.rollNo}</td>
<td>{student.name}</td>
<td>{student.course}</td>
<td>{student.marks}</td>
</tr>
))}
</tbody>
</table>
</div>
);
}
export default App;
```

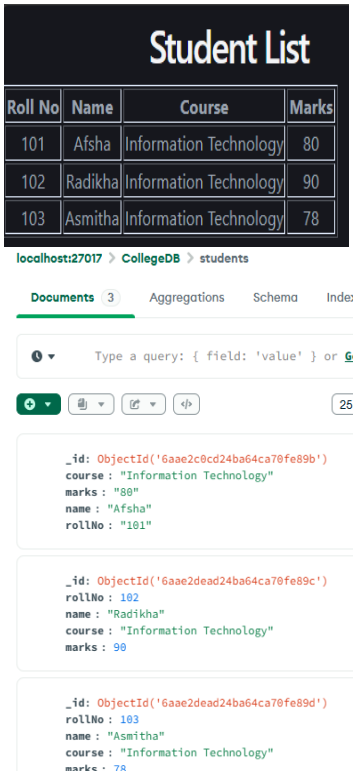
- server.js:

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
const app = express();
const PORT = 3000;
// Middleware
app.use(cors());
app.use(express.json());
// Connect MongoDB
mongoose.connect("mongodb://127.0.0.1:2701
7/CollegeDB")
.then(() => {
console.log("MongoDB Connected
Successfully");
})
.catch((error) => {
console.log("MongoDB Connection Failed");
console.log(error);
});
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app.get("/", (req, res) => {
res.send("Student Management API is
Running");
});
app.get("/students", async (req, res) => {
try {
const students = await
mongoose.connection.db
.collection("students")
.find()
.toArray();
res.json(students);
}
catch (error) {
res.status(500).json({
message: "Error fetching students"
});
}
});
app.listen(PORT, () => {
console.log(`Server is running on
http://localhost:${PORT}`);
});
```

Terminal command to run:

- node server.js
- npm run dev , if error then: dir
- Then command : npm run dev

Output:



D. Send data from React to the server using a POST request.
E. Store data in MongoDB.
F. Display stored data in the React page.
Code: App.jsx

```
import { useEffect, useState } from "react";
import axios from "axios";
function App() {
const [students, setStudents] = useState([]);
const [rollNo, setRollNo] = useState("");
const [name, setName] = useState("");
const [course, setCourse] = useState("");
const [marks, setMarks] = useState("");
// Get students
const getStudents = () =>
{axios.get("http://localhost:3000/students")
.then((response) =>
{setStudents(response.data);})
.catch((error) => {console.log(error);});
};
useEffect(() => {getStudents();}, []);
// Add student
const addStudent = () => {const newStudent =
{
rollNo: Number(rollNo),
name: name,
course: course,
marks: Number(marks)
};
axios.post("http://localhost:3000/students",
newStudent)
.then((response) =>
{alert(response.data.message);
setRollNo("");
setName("");
setCourse("");
setMarks("");
getStudents();
})
.catch((error) => {
console.log(error);
});
};
return (
<div>
<h1>Student Management System</h1>
<h2>Add Student</h2>
<input
type="number"
placeholder="Roll No"
value={rollNo}
onChange={(e) => setRollNo(e.target.value)}
/>
<br /><br />
<input
type="text"
placeholder="Student Name"
value={name}
onChange={(e) => setName(e.target.value)}
/>
<br /><br />
<input
type="text"
placeholder="Course"
value={course}
onChange={(e) => setCourse(e.target.value)}
/>
<br /><br />
<input
type="number"
placeholder="Marks"
value={marks}
onChange={(e) => setMarks(e.target.value)}
/>
<br /><br />
<button onClick={addStudent}>
Add Student
</button>
<hr />
<h2>Student List</h2>
<table border="1">
<thead>
<tr>
<th>Roll No</th>
<th>Name</th>
<th>Course</th>
```

```
<th>Marks</th>
</tr>
</thead>
<tbody>
{students.map((student) => (
<tr key={student._id}>
<td>{student.rollNo}</td>
<td>{student.name}</td>
<td>{student.course}</td>
<td>{student.marks}</td>
</tr>
))}
</tbody>
</table>
</div>
);
}
export default App;
```

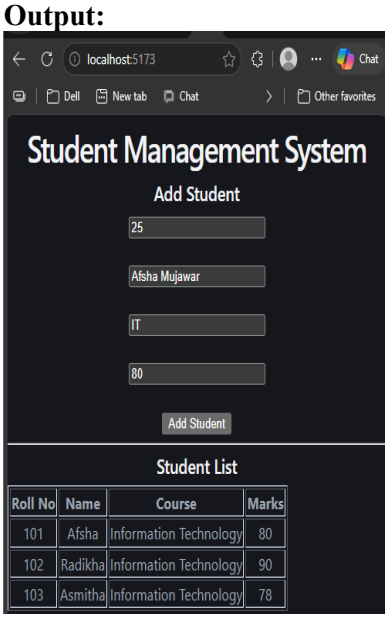
- server.js:

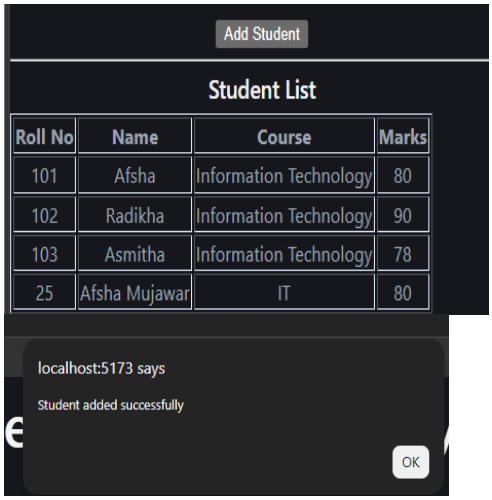
```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
const app = express();
const PORT = 3000;
// Middleware
app.use(cors());
app.use(express.json());
// Connect MongoDB
mongoose.connect("mongodb://127.0.0.1:27017/CollegeDB")
  .then(() => {
    console.log("MongoDB Connected Successfully");
  })
  .catch((error) => {
    console.log("MongoDB Connection Failed");
    console.log(error);
  });
// Simple REST API
app.get("/", (req, res) => {
  res.send("Student Management API is Running");
});
app.get("/students", async (req, res) => {
  try {
    const students = await mongoose.connection.db
      .collection("students")
      .find()
      .toArray();
    res.json(students);
  }
  catch (error) {
    res.status(500).json({
      message: "Error fetching students"
    });
  }
});
app.post("/students", async (req, res) => {
  try {
    const student = req.body;
    await mongoose.connection.db
      .collection("students")
      .insertOne(student);
    res.json({
      message: "Student added successfully",
      student: student
    });
  }
  catch (error) {
    res.status(500).json({
      message: "Error adding student"
    });
  }
});
app.listen(PORT, () => {
  console.log(`Server is running on http://localhost:${PORT}`);
});
```

Terminal command to run:

- node server.js
- npm run dev , if error then: dir
- Then command : npm run dev

Output:




Add Student
Student List
Roll No
Name
Course
Marks
101
Afsha
Information Technology
80
102
Radikha
Information Technology
90
103
Asmitha
Information Technology
78
25
Afsha Mujawar
IT
80
localhost:5173 says
Student added successfully
OK